

Investigating Agile Management Fashions at Rohde & Schwartz

Mindfully Navigating the Landscape of Agile Development Methods

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ABSTRACT

Adoption of agile development methods is becoming a natural challenge for software development organisations (SDO). Prior literature has proven how adoption may follow different levels, dabbler, tailor, or crusade. This present study conducts a case study of a self-proclaimed tailor SDO, and inquiries into how they adopt agile development methods. Results show that the development team shows a mindful approach in the adoption, and that reflections on the problem they have, and the proposed methods are carefully evaluated throughout the team. The results are discussed against Mintzberg's organisation configurations.

INTRODUCTION

We have seen a rise in agile development methods with the birth of the Agile Manifesto [1] two decades ago (e.g. SCRUM [11], XP[9] and Kanban[2]). With these methods follow agile techniques such as daily stand up meetings, retrospectives, pair programming and shared code ownership. In practice, Software Development Organisations (SDO) adopt agile methods, and thereby techniques, on different levels. Through this adoption, SDO's mindfully or mindlessly adopt and navigate within agile development methods, by either being a dabbler, tailor or crusader. Cram and Newell examine agile development as a management fashion[4] and concludes how SDO's demonstrate indicators of their commitment and adoption of a specific agile development method. Tailored SDO's do not commit to following agile development methods fully, as they exhibit flexibility in their utilisation of agile methods, by modifying methods and techniques based on the project at hand, and by mixing and integrating both agile and classical plan-driven development methods.

With the wide array of agile development and even more embedded techniques, tailored SDO's need to exhibit a thorough reflective practice[10] navigating the landscape of agile development. This navigation directly

influences tailored SDO's software development culture and in which they are at constant risk of either mindlessly succumbing to the promising fashions of agile development (crusading) or superficially adopting agile techniques (dabbling). Reflective tailors may be seen as inherently eclectic, as they select methods of what suits their organisation and meets their requirements within their development team.

Therefore, this paper aims at inquiring how the theory of agile software development meets a practice[8] of a self-proclaimed tailored SDO. Leading to the problem statement of this paper:

How does a company with an eclectic development methodology navigate and adopt agile development?

The remainder of this paper is structured as follows: First, agile development as a management fashion will be presented. Second, the research approach will follow with a description of the case in question and how the study was conducted. Next, the findings from the study will be presented with a management fashion lens. Finally, I discuss my findings against related work.

AGILE DEVELOPMENT AS A MANAGEMENT FASHION

To answer the problem statement, I use Cram and Newell's management fashion as the theoretical lens. Management fashion argues that there are transitory factors that influence the adoption of new techniques within organisations and whether these adoptions are conducted mindfully or mindlessly. In the mindful state, organisations are alert and open in complex situations, contrasting the mindless state where organisations ignore new information and exhibit a lack of attention to details.

Cram and Newell suggest five management fashion indicators that SDO's may exhibit when adopting agile development methods, being Norm of progress, Norm of rationality, Socio-psychological forces, Techno-economic

forces, and Innovation devolution (Table 1). The indicators cannot be characterised as good or bad but are set of indicators that enable an inquiry into mindful or mindless adoption of agile development methods.

Table 1. Management fashion indicators

Indicator	Relation to agile development
Norm of progress	Techniques are seen as novel
Norm of rationality	Pronouncement from others that they are proficient
Socio-psychological forces	Fulfills psychological needs of employees.
Techno-economic forces	Fitting with a company's collaborative culture
Innovation devolution	Ceremonially adopted that diverges from the intention

Finally, Cram and Newell present three categories (crusader, tailor and dabbler) of agile development adoption and use. Crusaders follow the prescriptions of the methods and implement the techniques consistently without mixing methods of other methodologies into their development process. Tailors will not blindly adhere to agile development methods, as they exhibit flexibility in their utilisation of agile methods. They modify methods and techniques based on the project at hand and combine various development methodologies. Dabblers adopt a small subset of agile techniques and continue in large parts to follow a plan-driven methodology.

METHOD

The context of the case study is the RnD development team of 14 people at Rohde & Schwarz Technology Center in Aalborg (Table 2). Rohde & Schwartz (R&S) is a global provider of electronics equipment in Defence, Media, Cybersecurity and Testing [12]. The team is developing and designing software to be used either directly by R&S's electronic equipment or as utility software.

Table 2. Interviewee and organisational overview.

Rohde & Schwarz Technology Center	
Industry	Electronic equipment
Number of interviews	4
Interviewee roles	UX and development, Product Manager, Quality Assurance, Team Lead

Style of agile approach	SCRUM
Initial data of adoption	2014

For the current research, we conducted a holistic single case study [13], by conducting four semi-structured interviews [7]. The data was collected through audio recordings, transcribed and finally coded. The interview was thematically coded based on the management fashion indicators. With the inspiration of Ghobadi & Mathiassen, each interviewee role was coded independently [5], with respect to a differing perspective among the roles.

FINDINGS

In this section, I present the findings of the analysis categorized within the management fashion indicators. In the analysis it became apparent that the interviews were conducted with varying agenda and scope, ultimately inflicting the analysis and results. For instance, the word "agile" is within the UX interview only appear two times in one question. Therefore, the UX interview is disregarded from the findings. A total of 29 management fashion indicators was identified in the three interviews of the Product Manager (PM), Team Lead (TL) and Quality Assurance Engineer (QA) (Table 3).

Table 3. Frequency of observed indicators.

Indicators	PM	TL	QA	T
Norm of progress	1	1	0	2
Norm of rationality	0	2	0	3
Socio-psychological forces	1	3	0	4
Techno-economic forces	4	10	1	14
Innovation devolution	2	3	1	6
Total (T)	8	19	2	29

Norm of Progress

In the two instances of Norm of Progress, the PM and TL expressed how their years in the field have made them open to new novel techniques.

"Today you will giggle because of it (waterfall), because of the realization that it doesn't work [...] I'll not surprise me if we look back at SCRUM in x amount of years, and we all giggle about that" – TL

This shows how it is important to be open about new techniques, but just as well be able to look back on former practices. This point is further elaborated with:

"It moves so quickly (development methods), so it is important to combine someone that knows how to do it (new development methods) and the people with the domain knowledge" - PM

Norm of Progress is instantiated in R&S by being open to novel techniques and be aware and mindful in the adoption process.

Norm of rationality

Norm of rationality emerged as the way they valued new introductions and suggestions of agile development methods from new or old colleagues. When they hire student straight of the universities, they actively have them review their methods to keep up with the newest agile development methods, here the TL figuratively asking a new hire: "So, now that you have been here for a while, what do you think?" with the rationale: "Or else we won't learn anything, and then we just become old and sullen".

At the same time the TL explains that an internal resistance against already tried out methods, TL makes a developer caricature: "I have found the Waterfall-model V3, anybody want to join?" and concludes: "It is a collective agreement that it doesn't work anymore [...] it might even happen to SCRUM one day". This shows how the teams are in a shared understanding of what they believe works the best, however they stay open to trying new methods with the recommendation of others.

Socio-psychological forces

Both the PM and TL stressed how important it is that the methods they follow allow the developers to be creative and free to experiment. This is upheld through designated process days:

"What works and don't? [...] and then decide democratically what we like, and what parts we don't like"

Ultimately, the quotes point at the importance of having agile development methods as a gain to the developer team's psychological needs and not as a means for having designated processes.

Techno-economic forces

In the three interviews, the importance of agility within the developer team was stressed as an important part of their work processes. This is in strong contrast with the mother company, where they manufacture through the production line, and where their relations with large

public customers are plan-driven and leaves little room for agile practices. As stated with:

"From the outside, we may seem waterfall-driven [...] and R&S is a conventional firm that thinks in waterfall. We try to internally navigate in an agile manner" - QA

The agile practices in the development team also support the collaborative culture, as it provides techniques for sharing accomplishments and challenges for instance through a Friday meeting, where they present current statuses on the project and even technological musings that have emerged through their work.

Innovation devolution

In many aspects of the interviews, R&S employed specific techniques from agile methods. However, contrary to Cram & Newell's Innovation devolution they did it at the aim of understanding what would best fit their development team. This is highlighted by an iterative approach to the agile development methods where the team experiences problems in their situation in which they aim to solve by tailoring methods to fit. The method that is adopted, are adopted at a very concrete level with a very specific goal in mind. Encapsulated in this quote from the TL:

"You always need to remember that a process is fantastic in its prescription, but it is not creating any value in itself" - TL

Highlighting that although R&S diverges from intentions from the agile method creators, they do so in a very reflective and mindful manner.

DISCUSSION

In relation to Mintzberg organisation configurations, R&S Technology Centre exhibits behaviours of an Innovative organisation with large amounts of mutual adjustment [6]. Mutual adjustment in the sense that they adjust independently within the development team and in that they have autonomy of what methods they may implement, ultimately pulling towards collaboration. As stated by Mintzberg companies need to decentralise the structure in order to realise innovation, and R&S does so by selected decentralisation. A higher level of innovation may even be reached by further collaboration and knowledge-sharing with more business units, customers and other stakeholders [3].

The development team in RnD exhibit how they share achievements and challenges at their Friday meetings, but they also conduct informal technological meetings for alignment and knowledge sharing, which results in a small organisation with limited knowledge sharing barriers [5]. However, with the physical decentralisation barriers of project organisation and project setting do emerge.

Mindful tailoring

My problem statement was: *How does a company with an eclectic development methodology navigate and adopt agile development?* Through these interviews it shows how R&S has been through a journey of five years modifying, replacing and adopting various agile development methods. It is clear that they conduct a very reflective practice of both selecting and testing out methods, and by leveraging their development teams experience they enable themselves to mindfully adopt new agile methods and techniques that solve a current development practice situation. The development team, and especially the TL, exhibit a highly pragmatic way of thought while adopting agile development methods, which is indeed mindful tailoring.

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